

Manila Barangay Flood Risk Priority Classification Report

Disaster Prioritization Index (DPI) and Machine Learning Classification-Reproduction Results

Study Area: City of Manila, Philippines

Unit of Analysis: Barangay-level total enumeration

Expected Barangay Count: 897 barangays

Main Outputs: Disaster Prioritization Index, Low/Medium/High priority classes, machine learning model comparison, barangay ranking table, diagnostic audit tables, and feature importance results.

Generated: 2026-09-26 09:48:23

1. Project Overview

This report summarizes the outputs of the Manila barangay-level flood risk priority classification pipeline. The study covers all 897 barangays of the City of Manila and integrates LiPAD 5-year and 25-year flood hazard layers, PSA 2020 and PSA 2024 population data, barangay administrative information, and DEM-derived elevation.

The 100-year flood layer is excluded from the main Disaster Prioritization Index and machine learning feature set because it failed return-period consistency checks. The Disaster Prioritization Index combines three major components: Hazard, Exposure, and Vulnerability. Component scores are normalized and combined using entropy-derived weights.

The machine learning stage is interpreted as a classification-reproduction experiment. The target variable is the DPI-derived risk class. Therefore, model performance indicates how well selected raw and engineered features reproduce the DPI-derived priority structure. It should not be interpreted as independent prediction of actual flood events, flood damage, casualties, or future flood conditions.

2. Methodological Summary

Spatial Processing: Flood polygons are intersected with barangay boundaries to compute flood area and flood percentage for 5-year and 25-year return-period scenarios. Area-related calculations are performed using EPSG:32651.

Population Integration: PSA 2020 and PSA 2024 barangay-level population values are integrated. Population change and population density per hectare are computed for vulnerability assessment and machine learning.

Affected Population: Affected population is estimated by multiplying barangay population by flood percentage. This assumes uniform population distribution within each barangay.

DPI Computation: Hazard, Exposure, and Vulnerability component scores are computed from normalized indicators. Entropy weighting is then applied to determine final component weights. DPI scores are scaled from 0 to 100 and classified into Low, Medium, and High priority groups using tertiles.

3. Exposure-Weight Interpretation

The entropy-weighted DPI may assign a larger weight to the Exposure component when affected-population values vary more strongly across barangays than the Hazard or Vulnerability components. This does not mean that hazard or vulnerability indicators are unimportant. Instead, it means that, within the Manila barangay dataset, differences in potentially affected population are a major source of variation in final priority scores.

As a result, some highly populated barangays may rank highly even if their flood coverage percentage is lower than that of smaller barangays. This is consistent with a disaster-prioritization objective, where the number of potentially affected residents is an important consideration for preparedness planning, monitoring, and resource allocation.

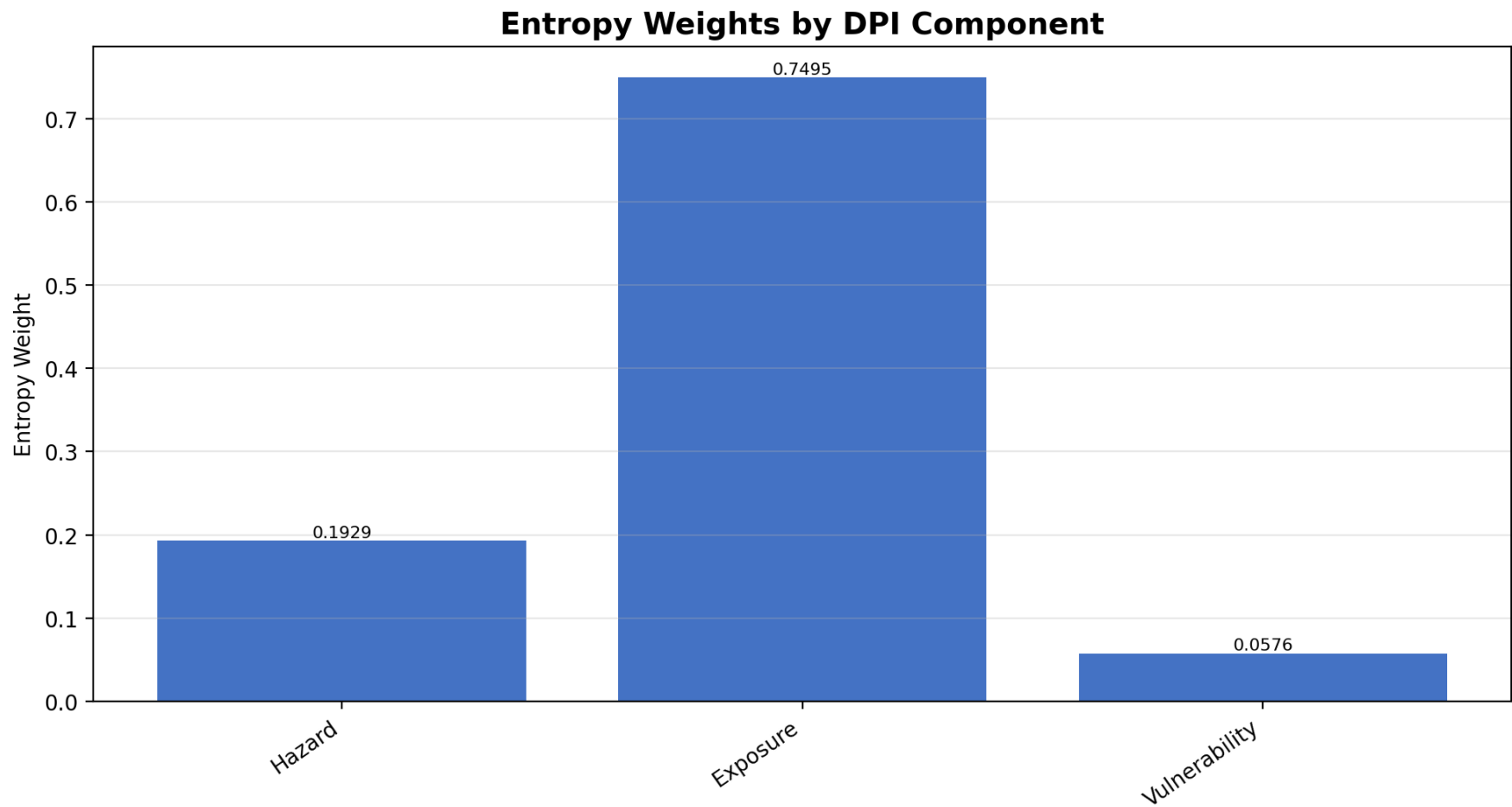
4. Entropy Component Weights

Entropy weights describe the data-driven contribution of Hazard, Exposure, and Vulnerability to the final DPI.

Component	Entropy_ Value	Diversification_ Value	Entropy_ Weight
Hazard	0.9877	0.0123	0.1929
Exposure	0.9521	0.0479	0.7495
Vulnerability	0.9963	0.0037	0.0576

Entropy Weights by DPI Component

Higher entropy weight indicates greater component contribution based on data dispersion.



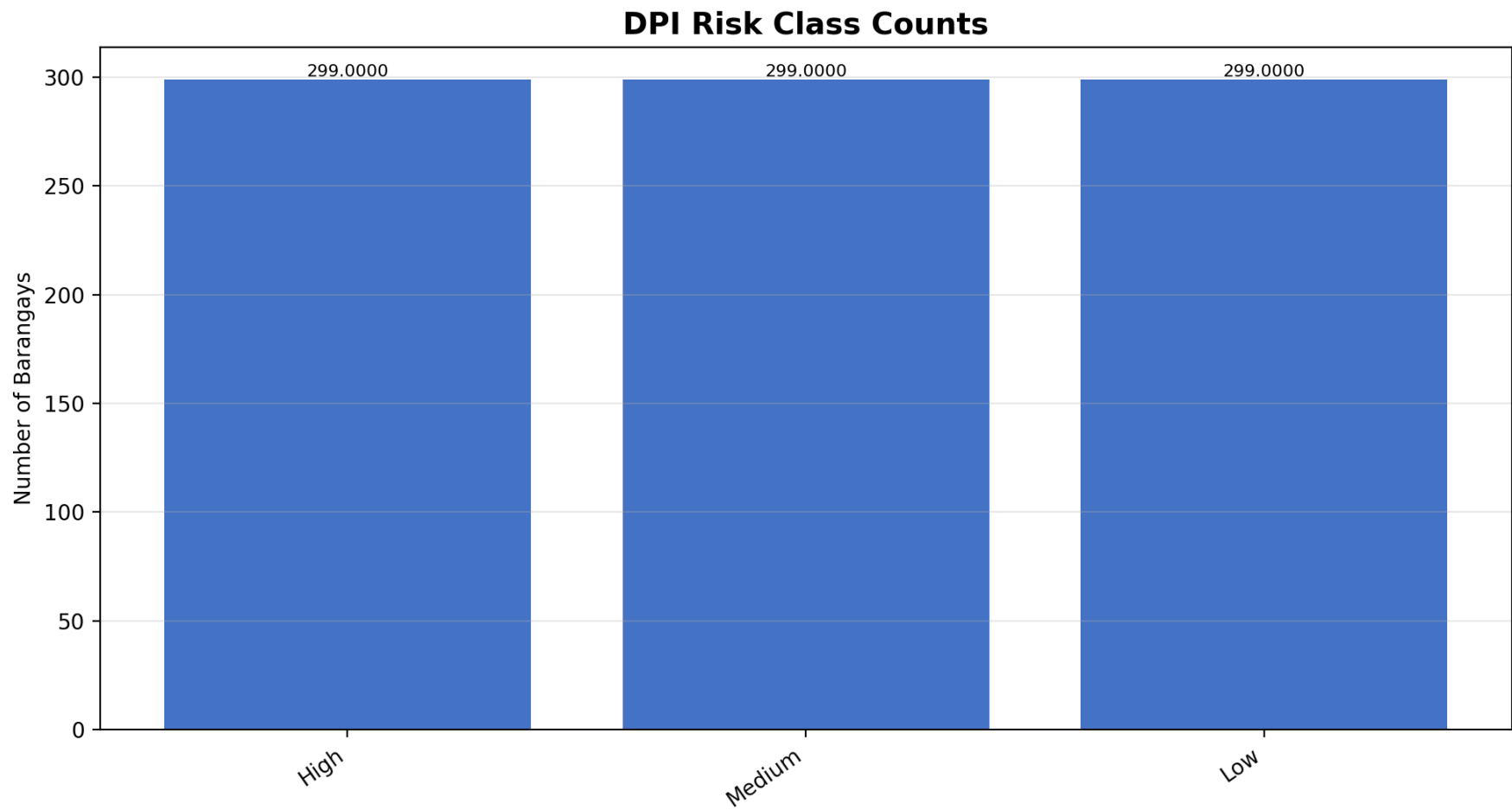
5. DPI Risk Class Distribution

Tertile classification produces equal Low, Medium, and High priority groups.

DPI_Risk_Class	Barangay_Count
High	299
Medium	299
Low	299

DPI Risk Class Counts

The classes are relative priority tertiles within Manila.



6. DPI Component and Class Summary

Wide tables are split into smaller sub-tables to improve readability.

DPI and Count

DPI_Risk_Class	Barangay_Count	Mean_DPI	Min_DPI	Max_DPI
Low	299	11.6847	0.0000	17.6842
Medium	299	20.2100	17.6946	22.8888
High	299	30.2358	22.9297	100.0000

Component and Hazard Summary

DPI_Risk_Class	Mean_Exposure	Mean_Vulnerability	Mean_Flood_25yr
Low	0.0336	0.3143	51.2254
Medium	0.0644	0.3351	94.3240
High	0.1637	0.3750	92.7015

Population and Elevation Summary

DPI_Risk_Class	Mean_Population_2024	Mean_Elevation
Low	1584.0736	2.8729
Medium	1234.9130	2.1278
High	3544.1906	1.9839

7. Top-Ranked Barangays by DPI

High-ranking barangays may be driven by flood coverage, high exposed population, low elevation, density, or combined factors.

Identification and Ranking

Barangay	Barangay_ No	DPI_ Risk_ Class	DPI_ Scaled
Barangay 310	310	High	100.0000
Barangay 628	628	High	98.3599
Barangay 649	649	High	78.7065
Barangay 598	598	High	71.2785
Barangay 648	648	High	67.8808
Barangay 20	20	High	67.5424
Barangay 843	843	High	66.6013
Barangay 900	900	High	66.2743
Barangay 397	397	High	64.5907
Barangay 898	898	High	55.1067
Barangay 101	101	High	53.9209
Barangay 152	152	High	48.3606
Barangay 836	836	High	46.6055
Barangay 830	830	High	45.9527
Barangay 128	128	High	45.8897
Barangay 183	183	High	44.2821
Barangay 630	630	High	42.6757
Barangay 275	275	High	42.4574
Barangay 781	781	High	42.1833
Barangay 116	116	High	41.8591
Barangay 48	48	High	40.0222
Barangay 129	129	High	39.3909
Barangay 829	829	High	39.1214
Barangay 595	595	High	38.7884

Barangay	Barangay_ No	DPI_ Risk_ Class	DPI_ Scaled
Barangay 396	396	High	38.7034

Population and Flood Exposure

Barangay	Population_ 2024	Flood_ PCT_ 25yr	Affected_ Population_ 25yr
Barangay 310	13761.0000	98.9598	13617.8562
Barangay 628	16279.0000	85.6499	13942.9548
Barangay 649	69769.0000	17.4492	12174.0977
Barangay 598	10510.0000	100.0000	10510.0000
Barangay 648	9086.0000	97.3978	8849.5669
Barangay 20	43691.0000	23.6280	10323.3138
Barangay 843	9355.0000	100.0000	9355.0000
Barangay 900	8689.0000	96.8465	8414.9882
Barangay 397	8835.0000	99.7573	8813.5604
Barangay 898	7812.0000	91.8035	7171.6896
Barangay 101	13659.0000	55.1735	7536.1495
Barangay 152	6016.0000	100.0000	6016.0000
Barangay 836	5673.0000	97.6192	5537.9382
Barangay 830	5407.0000	94.9908	5136.1512
Barangay 128	22298.0000	30.9103	6892.3753
Barangay 183	8512.0000	69.7958	5941.0208
Barangay 630	6552.0000	76.5651	5016.5460
Barangay 275	20995.0000	29.2805	6147.4448
Barangay 781	4337.0000	98.4771	4270.9513
Barangay 116	6182.0000	80.0858	4950.9028
Barangay 48	4191.0000	99.9876	4190.4805
Barangay 129	5207.0000	85.1279	4432.6080
Barangay 829	4375.0000	91.3555	3996.8028

Barangay	Population_2024	Flood_PCT_25yr	Affected_Population_25yr
Barangay 595	4229.0000	95.0682	4020.4355
Barangay 396	4237.0000	100.0000	4237.0000

Component Scores and Elevation

Barangay	Elevation_Mean	Hazard	Exposure	Vulnerability
Barangay 310	2.0000	0.6135	0.8478	0.4277
Barangay 628	1.4089	0.5684	0.8452	0.3930
Barangay 649	0.6252	0.2207	0.7316	0.4007
Barangay 598	2.5672	0.5000	0.5846	0.3820
Barangay 648	1.2499	0.5339	0.5296	0.5299
Barangay 20	1.2230	0.2753	0.6067	0.3469
Barangay 843	1.4798	0.5000	0.5204	0.5911
Barangay 900	1.9523	0.5639	0.5147	0.4081
Barangay 397	2.0000	0.5130	0.5014	0.5255
Barangay 898	1.8448	0.4928	0.4177	0.4104
Barangay 101	1.9019	0.3225	0.4499	0.4024
Barangay 152	1.0000	0.5028	0.3365	0.5291
Barangay 836	0.9785	0.5189	0.3220	0.4294
Barangay 830	1.1642	0.6173	0.2970	0.3372
Barangay 128	9.1411	0.2929	0.4020	0.0484
Barangay 183	2.0619	0.3574	0.3368	0.4653
Barangay 630	2.3527	0.4949	0.2959	0.3215
Barangay 275	1.3655	0.1973	0.3654	0.3851
Barangay 781	1.0000	0.5571	0.2656	0.4418
Barangay 116	3.0000	0.4403	0.2970	0.3818
Barangay 48	0.9083	0.5212	0.2448	0.5429
Barangay 129	2.4637	0.4694	0.2653	0.3654
Barangay 829	1.3646	0.5699	0.2386	0.3405

Barangay	Elevation_ Mean	Hazard	Exposure	Vulnerability
Barangay 595	2.1544	0.5325	0.2432	0.3615
Barangay 396	2.0000	0.5139	0.2407	0.4445

8. High-DPI but Moderate-Flood Audit

These barangays are classified as High priority despite below-median 25-year flood percentage. This often reflects high exposed population or vulnerability factors.

Showing first 25 rows for readability.

Identification and DPI

Barangay	DPI_ Risk_ Class	DPI_ Scaled
Barangay 628	High	98.3599
Barangay 649	High	78.7065
Barangay 20	High	67.5424
Barangay 898	High	55.1067
Barangay 101	High	53.9209
Barangay 830	High	45.9527
Barangay 128	High	45.8897
Barangay 183	High	44.2821
Barangay 630	High	42.6757
Barangay 275	High	42.4574
Barangay 116	High	41.8591
Barangay 129	High	39.3909
Barangay 829	High	39.1214
Barangay 595	High	38.7884
Barangay 105	High	38.2416
Barangay 601	High	37.9603
Barangay 905	High	37.7803
Barangay 91	High	37.2535
Barangay 719	High	37.1487
Barangay 739	High	36.7079
Barangay 659-A	High	34.5854
Barangay 776	High	34.1078
Barangay 103	High	33.9603

Barangay	DPI_ Risk_ Class	DPI_ Scaled
Barangay 831	High	33.8757
Barangay 704	High	33.7632

Flood and Exposure

Barangay	Population_ 2024	Flood_ PCT_ 25yr	Affected_ Population_ 25yr	Exposure
Barangay 628	16279.0000	85.6499	13942.9548	0.8452
Barangay 649	69769.0000	17.4492	12174.0977	0.7316
Barangay 20	43691.0000	23.6280	10323.3138	0.6067
Barangay 898	7812.0000	91.8035	7171.6896	0.4177
Barangay 101	13659.0000	55.1735	7536.1495	0.4499
Barangay 830	5407.0000	94.9908	5136.1512	0.2970
Barangay 128	22298.0000	30.9103	6892.3753	0.4020
Barangay 183	8512.0000	69.7958	5941.0208	0.3368
Barangay 630	6552.0000	76.5651	5016.5460	0.2959
Barangay 275	20995.0000	29.2805	6147.4448	0.3654
Barangay 116	6182.0000	80.0858	4950.9028	0.2970
Barangay 129	5207.0000	85.1279	4432.6080	0.2653
Barangay 829	4375.0000	91.3555	3996.8028	0.2386
Barangay 595	4229.0000	95.0682	4020.4355	0.2432
Barangay 105	26494.0000	21.4929	5694.3204	0.3346
Barangay 601	7765.0000	65.0607	5051.9644	0.2885
Barangay 905	7428.0000	62.4076	4635.6398	0.2743
Barangay 91	4453.0000	83.3521	3711.6677	0.2332
Barangay 719	8846.0000	51.7499	4577.7977	0.2684
Barangay 739	4598.0000	85.3194	3922.9842	0.2382
Barangay 659-A	3547.0000	94.2521	3343.1206	0.1916
Barangay 776	4606.0000	72.9258	3358.9607	0.2154

Barangay	Population_2024	Flood_PCT_25yr	Affected_Population_25yr	Exposure
Barangay 103	5071.0000	70.4801	3574.0464	0.2245
Barangay 831	3678.0000	91.7998	3376.3978	0.1958
Barangay 704	5339.0000	68.7056	3668.1934	0.2243

Vulnerability Context

Barangay	Elevation_Mean	Hazard	Vulnerability
Barangay 628	1.4089	0.5684	0.3930
Barangay 649	0.6252	0.2207	0.4007
Barangay 20	1.2230	0.2753	0.3469
Barangay 898	1.8448	0.4928	0.4104
Barangay 101	1.9019	0.3225	0.4024
Barangay 830	1.1642	0.6173	0.3372
Barangay 128	9.1411	0.2929	0.0484
Barangay 183	2.0619	0.3574	0.4653
Barangay 630	2.3527	0.4949	0.3215
Barangay 275	1.3655	0.1973	0.3851
Barangay 116	3.0000	0.4403	0.3818
Barangay 129	2.4637	0.4694	0.3654
Barangay 829	1.3646	0.5699	0.3405
Barangay 595	2.1544	0.5325	0.3615
Barangay 105	2.0458	0.1621	0.3389
Barangay 601	3.9096	0.3472	0.2803
Barangay 905	1.7744	0.3775	0.3406
Barangay 91	0.9116	0.4984	0.3986
Barangay 719	2.3433	0.3836	0.3114
Barangay 739	4.0000	0.4656	0.3706
Barangay 659-A	0.8715	0.5648	0.3603
Barangay 776	1.0000	0.4255	0.4531

Barangay	Elevation_ Mean	Hazard	Vulnerability
Barangay 103	3.0000	0.4032	0.3893
Barangay 831	1.3377	0.5263	0.3405
Barangay 704	1.6383	0.3833	0.4322

9. Elevation Outlier Audit

Elevation outliers should be retained unless confirmed as data errors, but they should be interpreted carefully in relation to exposure and vulnerability.

Showing first 20 rows for readability.

Elevation Statistics

Barangay	Barangay_ No	Elevation_ Mean	Elevation_ Median	Elevation_ Min	Elevation_ Max	Elevation_ Count
Barangay 128	128	9.1411	5.0000	0	30	11,552
Barangay 372	372	8.7610	9.0000	2	14	21,771
Barangay 191	191	8.6571	9.0000	8	9	420
Barangay 188	188	8.5180	9.0000	7	9	556
Barangay 189	189	8.2056	8.0000	7	9	968
Barangay 192	192	7.7603	8.0000	5	9	1,360
Barangay 585	585	7.7433	8.0000	5	10	2,723
Barangay 190	190	7.4334	8.0000	5	9	623
Barangay 587	587	7.2880	7.0000	4	13	2,937
Barangay 581	581	6.8011	7.0000	2	10	7,226
Barangay 187	187	6.7671	7.0000	5	9	438
Barangay 193	193	6.7610	7.0000	3	8	456
Barangay 194	194	6.3504	7.0000	4	8	682
Barangay 586	586	6.0671	6.0000	2	10	4,413
Barangay 540	540	6.0000	6.0000	6	6	885
Barangay 541	541	5.8852	6.0000	5	6	549
Barangay 555	555	5.8791	6.0000	5	7	695
Barangay 534	534	5.8526	6.0000	5	6	699
Barangay 589	589	5.6400	6.0000	3	9	1,314
Barangay 600	600	5.5650	5.0000	1	10	2,547

Flood and Population Context

Barangay	Flood_ PCT_ 25yr	Population_ 2024	Affected_ Population_ 25yr
Barangay 128	30.9103	22298.0000	6892.3753
Barangay 372	37.1815	4332.0000	1610.7024
Barangay 191	2.9185	754.0000	22.0054
Barangay 188	0.0000	872.0000	0.0000
Barangay 189	10.8450	1618.0000	175.4726
Barangay 192	22.5260	1771.0000	398.9351
Barangay 585	22.2620	1138.0000	253.3416
Barangay 190	0.8848	1282.0000	11.3433
Barangay 587	4.2991	2685.0000	115.4319
Barangay 581	8.9120	2212.0000	197.1340
Barangay 187	60.4579	717.0000	433.4831
Barangay 193	2.4230	928.0000	22.4855
Barangay 194	8.9595	746.0000	66.8379
Barangay 586	14.1002	6859.0000	967.1325
Barangay 540	22.4037	2024.0000	453.4506
Barangay 541	4.8700	812.0000	39.5447
Barangay 555	27.1593	1151.0000	312.6035
Barangay 534	12.3908	1128.0000	139.7680
Barangay 589	10.1958	559.0000	56.9948
Barangay 600	31.8069	6276.0000	1996.2013

DPI Context

Barangay	Vulnerability	Exposure	DPI_ Scaled	DPI_ Risk_ Class
Barangay 128	0.0484	0.4020	45.8897	High
Barangay 372	0.0563	0.0939	15.5540	Low
Barangay 191	0.1149	0.0013	0.5648	Low
Barangay 188	0.1090	0.0000	0.0000	Low

Barangay	Vulnerability	Exposure	DPI_Scaled	DPI_Risk_Class
Barangay 189	0.1279	0.0103	2.5728	Low
Barangay 192	0.1248	0.0230	5.2914	Low
Barangay 585	0.1006	0.0148	4.3956	Low
Barangay 190	0.1667	0.0006	0.6041	Low
Barangay 587	0.1281	0.0070	1.4504	Low
Barangay 581	0.1245	0.0117	2.6365	Low
Barangay 187	0.1658	0.0247	10.4981	Low
Barangay 193	0.1866	0.0013	1.0304	Low
Barangay 194	0.1704	0.0038	1.9565	Low
Barangay 586	0.2138	0.0592	8.7131	Low
Barangay 540	0.2263	0.0273	6.5373	Low
Barangay 541	0.2331	0.0025	1.8590	Low
Barangay 555	0.2087	0.0184	6.0797	Low
Barangay 534	0.2089	0.0081	3.1397	Low
Barangay 589	0.1684	0.0037	2.3030	Low
Barangay 600	0.2496	0.1179	16.8673	Low

10. Population-Change Sensitivity Audit

This diagnostic checks whether extreme population-change percentages materially affect the Vulnerability component.

Showing first 25 rows for readability.

Population Change

Barangay	Population_ 2020	Population_ 2024	Population_ Change_ 2020_ 2024_ Pct	Population_ Change_ 2020_ 2024_ Pct_ Winsorized
Barangay 481	1,004	1943.0000	93.5259	93.5259
Barangay 124	159	307.0000	93.0818	93.0818
Barangay 729	871	1775.0000	103.7887	93.9364
Barangay 8	147	301.0000	104.7619	93.9364
Barangay 271	633	1315.0000	107.7409	93.9364
Barangay 296	1,195	2273.0000	90.2092	90.2092
Barangay 262	1,736	3638.0000	109.5622	93.9364
Barangay 397	4,175	8835.0000	111.6168	93.9364
Barangay 692	1,329	2508.0000	88.7133	88.7133
Barangay 490	471	1048.0000	122.5053	93.9364
Barangay 632	254	576.0000	126.7717	93.9364
Barangay 697	1,967	3568.0000	81.3930	81.3930
Barangay 396	1,666	4237.0000	154.3217	93.9364
Barangay 676	1,431	2564.0000	79.1754	79.1754
Barangay 404	723	1268.0000	75.3804	75.3804
Barangay 696	2,420	4121.0000	70.2893	70.2893
Barangay 465	762	1292.0000	69.5538	69.5538
Barangay 407	1,571	2637.0000	67.8549	67.8549
Barangay 541	485	812.0000	67.4227	67.4227
Barangay 240	56	91.0000	62.5000	62.5000
Barangay 648	5,634	9086.0000	61.2709	61.2709

Barangay	Population_ 2020	Population_ 2024	Population_ Change_ 2020_ 2024_ Pct	Population_ Change_ 2020_ 2024_ Pct_ Winsorized
Barangay 392	695	1115.0000	60.4317	60.4317
Barangay 321	240	384.0000	60.0000	60.0000
Barangay 727	1,790	2842.0000	58.7709	58.7709
Barangay 730	1,539	2443.0000	58.7394	58.7394

Vulnerability Sensitivity

Barangay	Vulnerability	Vulnerability_ WinsorizedPopChange	Vulnerability_ Sensitivity_ Diff
Barangay 481	0.4243	0.6793	0.2551
Barangay 124	0.3457	0.5999	0.2542
Barangay 729	0.3531	0.6044	0.2513
Barangay 8	0.3409	0.5918	0.2508
Barangay 271	0.3834	0.6328	0.2494
Barangay 296	0.4274	0.6761	0.2487
Barangay 262	0.4492	0.6978	0.2486
Barangay 397	0.5255	0.7731	0.2476
Barangay 692	0.3596	0.6054	0.2458
Barangay 490	0.3912	0.6337	0.2426
Barangay 632	0.3842	0.6248	0.2406
Barangay 697	0.3649	0.5966	0.2317
Barangay 396	0.4445	0.6723	0.2278
Barangay 676	0.3089	0.5363	0.2274
Barangay 404	0.3441	0.5642	0.2201
Barangay 696	0.3557	0.5659	0.2103
Barangay 465	0.4098	0.6187	0.2088
Barangay 407	0.3653	0.5708	0.2056
Barangay 541	0.2331	0.4378	0.2047

Barangay	Vulnerability	Vulnerability_ WinsorizedPopChange	Vulnerability_ Sensitivity_ Diff
Barangay 240	0.3652	0.5604	0.1952
Barangay 648	0.5299	0.7227	0.1929
Barangay 392	0.3536	0.5449	0.1912
Barangay 321	0.3466	0.5370	0.1904
Barangay 727	0.3238	0.5119	0.1880
Barangay 730	0.3121	0.5001	0.1880

DPI Context

Barangay	DPI_ Scaled	DPI_ Risk_ Class
Barangay 481	25.6478	High
Barangay 124	17.2791	Low
Barangay 729	16.8952	Low
Barangay 8	14.2546	Low
Barangay 271	24.2545	High
Barangay 296	27.1541	High
Barangay 262	35.6689	High
Barangay 397	64.5907	High
Barangay 692	20.2557	Medium
Barangay 490	20.8989	Medium
Barangay 632	16.6122	Low
Barangay 697	26.9584	High
Barangay 396	38.7034	High
Barangay 676	29.1650	High
Barangay 404	21.7977	Medium
Barangay 696	25.4003	High
Barangay 465	21.7110	Medium
Barangay 407	30.1770	High

Barangay	DPI_ Scaled	DPI_ Risk_ Class
Barangay 541	1.8590	Low
Barangay 240	15.4230	Low
Barangay 648	67.8808	High
Barangay 392	20.7624	Medium
Barangay 321	16.5292	Low
Barangay 727	18.1084	Medium
Barangay 730	10.6616	Low

11. Machine Learning Model Comparison

Model performance reflects classification-reproduction of DPI-derived risk classes, not independent prediction of real flood outcomes.

Cross-Validation Performance

Model	CV_ F1_ Macro_ Mean	CV_ F1_ Macro_ Std	Best_ Params
Gradient Boosting	0.9343	0.0222	{'model_ _learning_ rate': 0.1, 'model_ _max_ depth': 2, 'model_ _n_ estimators': 200}
Random Forest	0.9312	0.0297	{'model_ _max_ depth': None, 'model_ _min_ samples_ leaf': 1, 'model_ _n_ estimators': 200}
MLP	0.8817	0.0401	{'model_ _alpha': 0.0001, 'model_ _hidden_ layer_ sizes': (64, 32)}

Held-Out Test Performance

Model	Accuracy	Balanced_ Accuracy	F1_ Macro	F1_ Weighted	ROC_ AUC_ OVR_ Macro
Gradient Boosting	0.9167	0.9167	0.9168	0.9168	0.9863
Random Forest	0.9056	0.9056	0.9061	0.9061	0.9898

Model	Accuracy	Balanced_ Accuracy	F1_ Macro	F1_ Weighted	ROC_ AUC_ OVR_ Macro
MLP	0.9333	0.9333	0.9336	0.9336	0.9910

High-Class Performance

Model	Precision_ High	Recall_ High	F1_ High	Support_ High
Gradient Boosting	0.9333	0.9333	0.9333	60
Random Forest	0.9483	0.9167	0.9322	60
MLP	0.9500	0.9500	0.9500	60

Low-Class Performance

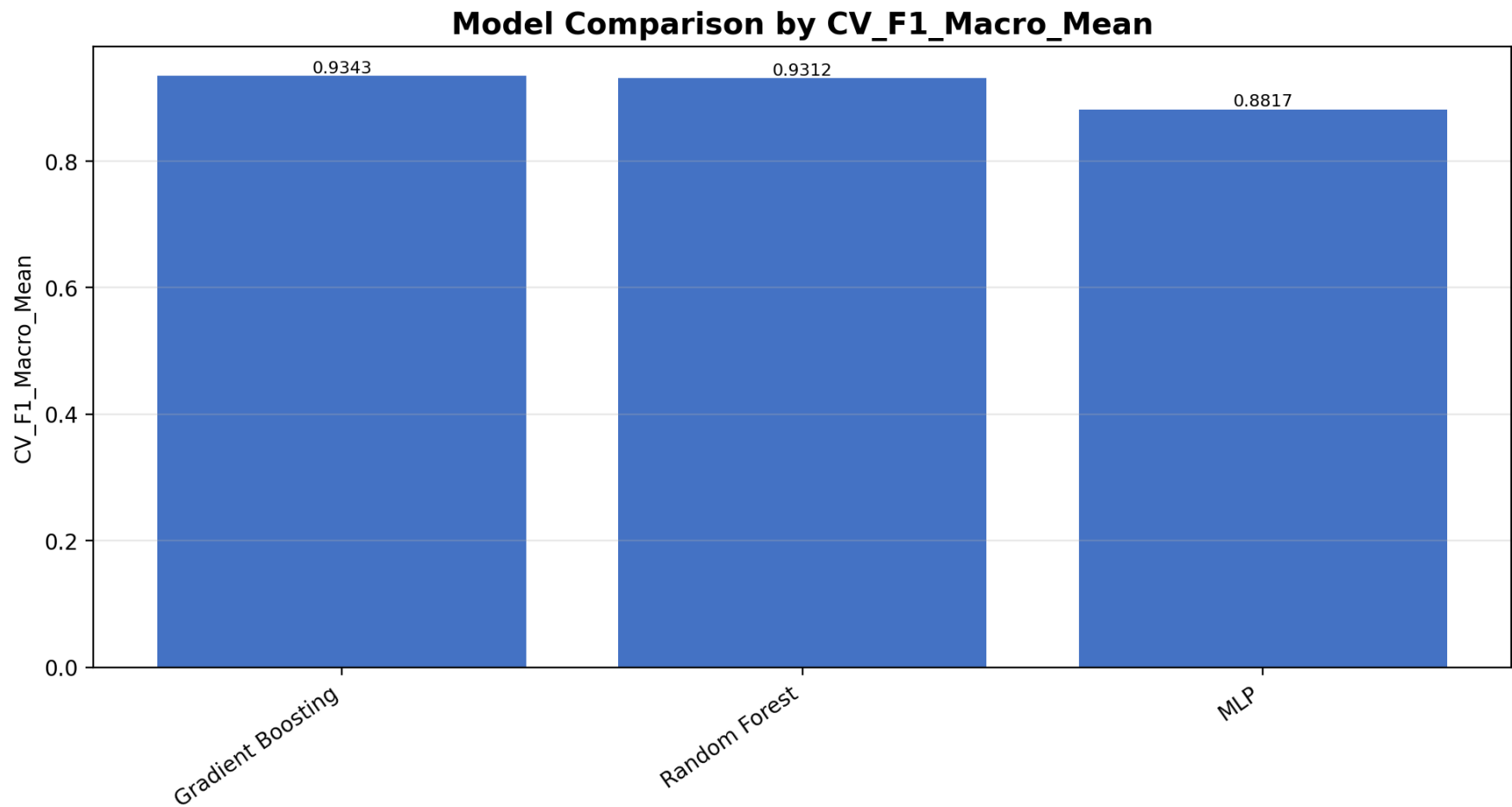
Model	Precision_ Low	Recall_ Low	F1_ Low	Support_ Low
Gradient Boosting	0.9492	0.9333	0.9412	60
Random Forest	0.9322	0.9167	0.9244	60
MLP	0.9655	0.9333	0.9492	60

Medium-Class Performance

Model	Precision_ Medium	Recall_ Medium	F1_ Medium	Support_ Medium
Gradient Boosting	0.8689	0.8833	0.8760	60
Random Forest	0.8413	0.8833	0.8618	60
MLP	0.8871	0.9167	0.9016	60

Model Comparison by CV_F1_Macro_Mean

The selected model should be based on the predefined model-selection metric, typically cross-validation F1-macro.



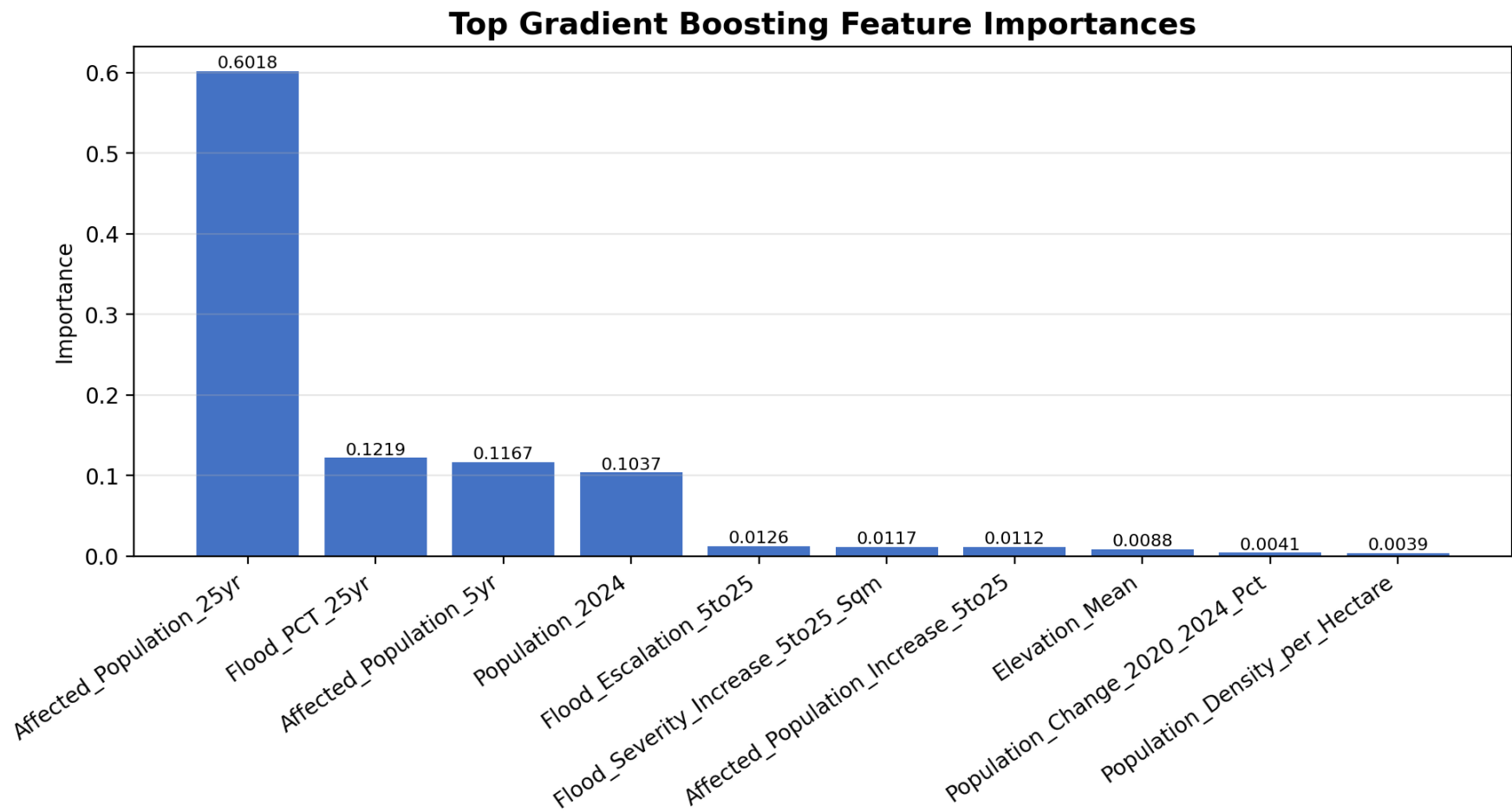
12. Gradient Boosting Feature Importance

Impurity-based feature importance explains how the Gradient Boosting model reproduced DPI-derived classes.

Feature	Importance	Model
Affected_Population_25yr	0.6018	Gradient Boosting
Flood_PCT_25yr	0.1219	Gradient Boosting
Affected_Population_5yr	0.1167	Gradient Boosting
Population_2024	0.1037	Gradient Boosting
Flood_Escalation_5to25	0.0126	Gradient Boosting
Flood_Severity_Increase_5to25_Sqm	0.0117	Gradient Boosting
Affected_Population_Increase_5to25	0.0112	Gradient Boosting
Elevation_Mean	0.0088	Gradient Boosting
Population_Change_2020_2024_Pct	0.0041	Gradient Boosting
Population_Density_per_Hectare	0.0039	Gradient Boosting
Flood_PCT_5yr	0.0036	Gradient Boosting

Top Gradient Boosting Feature Importances

Higher values indicate stronger contribution to the model's classification-reproduction behavior.



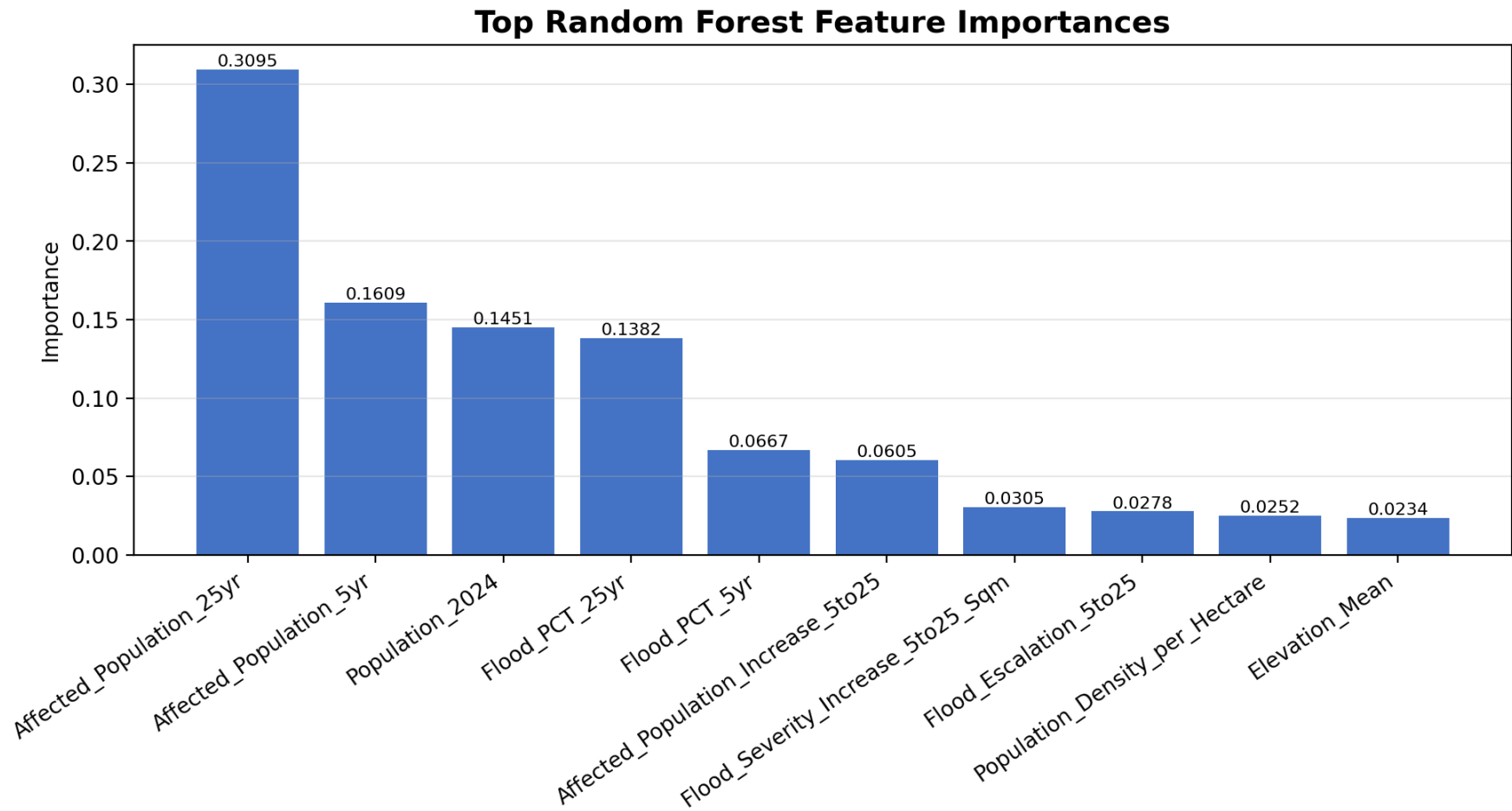
13. Random Forest Feature Importance

Impurity-based feature importance from the Random Forest model.

Feature	Importance	Model
Affected_Population_25yr	0.3095	Random Forest
Affected_Population_5yr	0.1609	Random Forest
Population_2024	0.1451	Random Forest
Flood_PCT_25yr	0.1382	Random Forest
Flood_PCT_5yr	0.0667	Random Forest
Affected_Population_Increase_5to25	0.0605	Random Forest
Flood_Severity_Increase_5to25_Sqm	0.0305	Random Forest
Flood_Escalation_5to25	0.0278	Random Forest
Population_Density_per_Hectare	0.0252	Random Forest
Elevation_Mean	0.0234	Random Forest
Population_Change_2020_2024_Pct	0.0121	Random Forest

Top Random Forest Feature Importances

Feature importance should be interpreted cautiously when predictors are correlated.



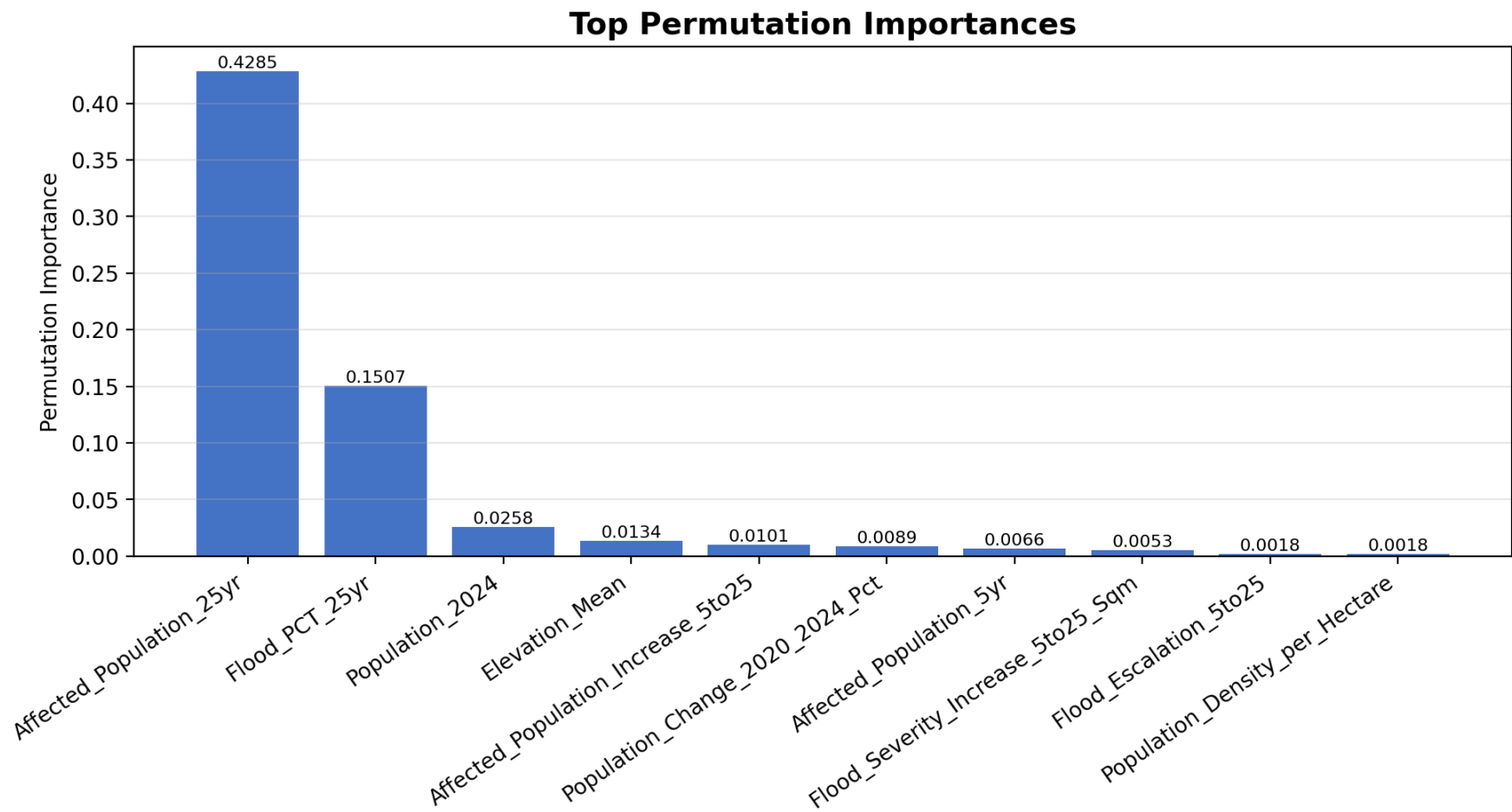
14. Permutation Importance

Permutation importance estimates how much each feature contributes to held-out F1-macro when feature values are randomly shuffled.

Feature	Permutation_Importance_Mean	Permutation_Importance_Std	Model
Affected_Population_25yr	0.4285	0.0358	Gradient Boosting
Flood_PCT_25yr	0.1507	0.0212	Gradient Boosting
Population_2024	0.0258	0.0107	Gradient Boosting
Elevation_Mean	0.0134	0.0078	Gradient Boosting
Affected_Population_Increase_5to25	0.0101	0.0102	Gradient Boosting
Population_Change_2020_2024_Pct	0.0089	0.0037	Gradient Boosting
Affected_Population_5yr	0.0066	0.0107	Gradient Boosting
Flood_Severity_Increase_5to25_Sqm	0.0053	0.0079	Gradient Boosting
Flood_Escalation_5to25	0.0018	0.0050	Gradient Boosting
Population_Density_per_Hectare	0.0018	0.0064	Gradient Boosting
Flood_PCT_5yr	0.0017	0.0045	Gradient Boosting

Top Permutation Importances

Permutation importance is model-agnostic and helps validate impurity-based feature-importance patterns.



15. Limitations and Usage Notes

1. The study relies on secondary datasets and does not include field validation, household surveys, or complete barangay-level historical damage records.
2. LiPAD flood layers represent modeled hazard scenarios and may not fully reflect real-time flooding, drainage constraints, rainfall intensity, tidal influence, pumping conditions, land-use changes, or recent infrastructure updates.
3. The 100-year flood layer is excluded from the main analysis due to return-period consistency anomalies. Any retained 100-year fields in intermediate files are for traceability only.
4. Affected population is estimated using barangay-level flood percentage and total population. This assumes a uniform distribution of residents within each barangay, which may not fully reflect actual settlement patterns.
5. Elevation is summarized at the barangay level and does not capture micro-scale road depressions, building elevation, local drainage obstructions, land subsidence, or flood depth.
6. Low, Medium, and High classes are relative tertiles within Manila. They are not official flood-warning categories or externally calibrated hazard levels.
7. Machine learning performance measures classification-reproduction of DPI-derived classes. The models do not independently predict actual flood occurrence, flood depth, damage, casualties, or future flood conditions.
8. The prototype and outputs are intended for research and planning support only. They should not replace official government hazard advisories, evacuation orders, or operational disaster response systems.